

Benchmarking Small Language Models for Domain-Specific Question Answering: A Comparative Study of Phi-3-mini, Mistral-7B, and Gemma-2 on SQuAD v2

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Abstract—Small Language Models (SLMs) offer competitive natural language understanding at a fraction of the compute cost of Large Language Models (LLMs). Despite this, few studies compare multiple SLMs on the same question answering (QA) benchmark under identical conditions. We evaluate three instruction-tuned SLMs, Phi-3-mini-4k-instruct (3.8B), Mistral-7B-Instruct-v0.2 (7B), and Gemma-2-2b-it (2B), on the SQuAD v2 dataset in a zero-shot setting. Each model is scored on Exact Match (EM), F1, and average inference time per query. All three models produce low EM and F1 scores, not because they fail to comprehend the passages, but because they generate full sentences rather than short answer spans. We examine this generative-extractive mismatch and discuss what it means for practitioners deploying SLMs in QA pipelines.

I. INTRODUCTION

Transformer-based language models have improved rapidly across a wide range of NLP tasks [1]. LLMs such as GPT-3 [2] perform well on many benchmarks, but their memory and compute requirements make them costly to run in production, particularly in low-resource or latency-sensitive settings [3].

SLMs address this by fitting on consumer-grade hardware [4]. Models like Phi-3-mini [5], Mistral-7B [6], and Gemma-2 [7] achieve competitive NLP scores at much lower inference cost. Belcak et al. [4] go further, arguing that SLMs are not just cheaper but better suited to the repetitive, scoped tasks that appear most often in deployed systems. What is missing from the literature is a direct comparison of these three models on the same QA benchmark, evaluated under the same conditions.

QA is a useful stress test for language models because it requires reading comprehension and, in the case of SQuAD v2, the ability to recognize when a question has no answer [8]. Brogly et al. [9] showed that Phi-3-mini’s published benchmark scores did not transfer cleanly to a real-world classification task, which raises the question of whether the same gap appears on QA.

This study makes three contributions:

- A controlled zero-shot comparison of Phi-3-mini-4k-instruct, Mistral-7B-Instruct-v0.2, and Gemma-2-2b-it on SQuAD v2.

- EM, F1, and per-query inference time measured identically across all three models.
- An explanation of why generative SLMs score low on extractive QA metrics, and what that means for real deployments.

II. RELATED WORK

A. LLM and SLM Landscape

Language modeling shifted from n-gram statistics to neural approaches through the 1990s and 2000s, and took a decisive step with the transformer architecture of Vaswani et al. [1]. Örpek et al. [3] compare LLMs and SLMs directly, finding that LLMs cover more ground but SLMs consume far less energy and require less hardware. Sakib et al. [10] provide a broader taxonomy of SLM design strategies, covering lightweight architectures such as DistilBERT and MobileBERT, compression techniques including pruning and quantization, and evaluation suites that include SQuAD as a key benchmark for efficient inference. Their survey confirms that SQuAD remains a standard testbed for measuring SLM reading comprehension under latency constraints.

B. SLMs for Specialized Tasks

Belcak et al. [4] make a strong case for SLMs in production agentic systems, noting that serving a 7B model costs 10 to 30 times less than serving a 70B model in latency, energy, and floating-point operations. Microsoft’s Phi-3 series shows that 3.8B parameters is enough to match much larger models on reasoning tasks [5], which is one reason we include it here.

C. Domain-Specific Evaluation of SLMs

Brogly et al. [9] ran Phi-3-mini-4k-instruct over nine million Canadian news headlines, asking it to score each headline for medical and sports-injury relevance. When they compared those scores against seven human raters, the Spearman correlation was 0.23 to 0.39 depending on filtering. The takeaway is clear: a model can score well on published benchmarks and still behave differently on a real task. That finding motivates the controlled evaluation we run here.

TABLE I
MODELS EVALUATED IN THIS STUDY

Model (HuggingFace ID)	Params	Developer	License
Phi-3-mini-4k-instruct	3.8B	Microsoft	MIT
Mistral-7B-Instruct-v0.2	7B	Mistral AI	Apache 2.0
Gemma-2-2b-it	2B	Google	Gemma ToU

All models loaded via `microsoft/`, `mistralai/`, and `google/` namespaces on HuggingFace Hub, instruction-tuned variants without fine-tuning.

D. Question Answering Benchmarks

SQuAD v2 [8] builds on the original Stanford QA dataset by adding roughly 53,775 unanswerable questions. A model that cannot say “I don’t know” will score poorly. This makes it a harder and more realistic testbed than the original SQuAD, and it remains the standard benchmark for extractive reading comprehension.

E. Extractive vs. Generative QA Models

Extractive QA models such as BERT-based readers are trained to predict the start and end token positions of an answer span within the input context [11]. Generative readers, by contrast, produce answers autoregressively from the full vocabulary without being constrained to the input text. Luo et al. [11] conduct a systematic comparison of both reader types across 12 QA datasets, finding that extractive readers outperform generative ones on short-context datasets while generative readers show an advantage on longer passages. These structural differences in how the two reader types produce answers are central to interpreting the low EM and F1 scores observed in our evaluation.

III. METHODOLOGY

A. Dataset

We use SQuAD v2 [8], which contains around 150,000 question-answer pairs drawn from Wikipedia. Approximately half are unanswerable. We randomly sample 200 examples from the official validation split with a fixed seed of 42 so that results are reproducible and each model sees exactly the same questions.

B. Models

Table I lists the three models. All are instruction-tuned variants loaded directly from HuggingFace without any additional fine-tuning. This setup tests what each model can do out of the box, not what it can do after being trained on QA data. All models run in `float16` to reduce GPU memory use without meaningfully changing output quality.

C. Evaluation Metrics

We report three metrics per model:

- **Exact Match (EM):** The share of predictions that match the ground truth exactly after lowercasing and removing punctuation. For unanswerable questions the expected answer is an empty string.

TABLE II
ZERO-SHOT PERFORMANCE ON SQuAD v2 (200 EXAMPLES, SEED=42)

Model	EM (%)	F1 (%)	Inf. Time (ms)
Phi-3-mini-4k-instruct	0.00	8.92	2,465.8
Mistral-7B-Instruct-v0.2	1.00	12.40	14,062.2
Gemma-2-2b-it	8.00	19.46	33,516.8

- **F1 Score:** Token-level overlap between the predicted and ground truth answer, averaged across all questions. Partial credit is given for partially correct answers.
- **Average Inference Time:** Mean wall-clock time in milliseconds per query across all 200 examples, recorded with Python’s `time` module and excluding tokenization.

D. Experimental Setup

All experiments run on Google Colaboratory with a single NVIDIA T4 GPU and 15 GB VRAM. Each model receives this prompt:

```
Context: {context}
Question: {question}
Answer:
```

Contexts are truncated to 1,024 tokens to avoid out-of-memory errors on long Wikipedia passages. Generation is capped at 50 new tokens using greedy decoding (`do_sample=False`) so results are deterministic. Software versions: HuggingFace `transformers` 4.43.0, `datasets` 4.0.0, `evaluate` 0.4.6. Phi-3-mini requires `attn_implementation="eager"` for compatibility with this version of `transformers`.

IV. RESULTS

A. Exact Match and F1 Scores

Table II shows the results. Gemma-2-2b-it scores highest on both EM (8.00%) and F1 (19.46%), followed by Mistral-7B (EM 1.00%, F1 12.40%) and Phi-3-mini (EM 0.00%, F1 8.92%).

B. Inference Time Analysis

Phi-3-mini is the fastest at 2,465.8 ms per query. Mistral-7B takes 14,062.2 ms, about 5.7 times slower. Gemma-2-2b-it is slowest at 33,516.8 ms per query despite having the fewest parameters. The Gemma and Mistral runtimes reflect partial CPU offloading during inference, since their weights exceeded the available GPU VRAM headroom on the T4.

C. Comparative Analysis

Gemma-2 scores best on QA accuracy but costs the most in inference time. Phi-3-mini is the fastest but scores lowest on EM. Mistral-7B falls between both in accuracy and speed. No model dominates on all three metrics. The right choice depends on whether the deployment prioritizes answer quality or response latency.

V. DISCUSSION

A. Generative-Extractive QA Gap

The low EM scores are not what they look like at first glance. Inspecting model outputs shows that all three models understand the passages reasonably well. The problem is output format. When the correct answer is “1347,” a model generates “The Black Death reached the Mediterranean in 1347,” which contains the right information but scores zero under Exact Match because of the surrounding sentence.

Instruction-tuned SLMs are trained to give helpful, complete responses. SQuAD v2 expects a short extracted span, sometimes a single word or date. These goals are in direct conflict under zero-shot prompting. Extractive models like fine-tuned BERT are trained to predict a start and end token within the passage. Generative models produce text from scratch. Applying extractive metrics to generative output will always understate how much the model actually understood. Kamalloo et al. [12] confirm this quantitatively, showing that Exact Match underestimates zero-shot generative model performance by a significant margin because lexical matching penalizes semantically correct answers that differ only in wording from the ground truth, not in meaning. This is the core finding of this study, and it applies to any similar evaluation setup.

B. Inference Efficiency Trade-offs

Phi-3-mini is fast and parameter-efficient. Its 3.8B weights fit comfortably on a T4, and it runs fully on GPU throughout. Mistral-7B and Gemma-2 both trigger partial CPU offloading, which explains the much longer inference times. The fact that Gemma-2 has only 2B parameters but still runs slower than Mistral-7B suggests its architecture or generation behavior interacts poorly with the offloading pattern on this hardware.

C. Limitations

The evaluation covers 200 examples, which is enough to show trends but not enough to give tight confidence intervals. All three models run zero-shot, so the scores reflect base capability rather than what the models could do with task-specific fine-tuning. Bucher and Martini [15] confirm this directly, showing that fine-tuned smaller models consistently and significantly outperform larger zero-shot prompted models across NLP tasks, which suggests the scores reported here represent a lower bound on what these models are capable of rather than their ceiling.

The generative-extractive mismatch compounds this further, since EM and F1 undercount real comprehension whenever a model produces a correct answer in sentence form rather than as a bare span. Future work should explore post-processing pipelines that extract the answer span from longer generated outputs before scoring. Mallick et al. [13] demonstrate that generative models can be adapted for extractive metrics by generating token index pointers rather than free text, and Li and Ji [14] show that constrained decoding using trie-based prefix trees can restrict decoder output to tokens present in the context on SQuAD directly, making both approaches technically feasible starting points. Beyond output format, the

evaluation metric itself warrants revisiting. Emerging work suggests LLM-as-a-judge evaluation aligns better with human judgment than EM/F1 for generative QA outputs [16], offering a more appropriate benchmark for this model class in future studies.

VI. CONCLUSION

We compared Phi-3-mini-4k-instruct, Mistral-7B-Instruct-v0.2, and Gemma-2-2b-it on SQuAD v2 under identical zero-shot conditions. Gemma-2 scored highest on EM and F1. Phi-3-mini was fastest. All three models scored low in absolute terms, not from poor reading comprehension, but because instruction-tuned generative models produce full sentences while SQuAD v2 expects short spans. Getting accurate scores from generative SLMs on extractive benchmarks requires either a constrained decoding strategy or a post-processing step to pull the answer out of the generated text. Neither was applied here, which is the main factor behind the low numbers.

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